

Exercise - Trust

MIR IT-Protection WS25

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- Research one of the following topics, apply and reflect the knowledge and terms you learned in the lectures
 - max 2-3 people per topic
- Reflect critically: who wants what, what are the advantages or disadvantages, where are the problems to be solved?
 - Use your innovation & research curious mind!
- Some topics have brand-new updates – find current sources!
- Include references to material you used and how you found it
 - academic paper, videos, news articles, websites, ...

Topic 1 - Untrustworthy AI

- “AI will kill us all”
- Find doomsayers and what are their arguments that soon AI will be very, VERY bad for humanity.
- What can go (or has already gone very) wrong when AI is widely used in IT systems?

Topic 2 - Trust in Android

- Google is planning to restrict “sideloading” in Android, the possibility to simply install ANY application
- News say, starting in 2026 Android apps must be from a certain certified developer → developers must obtain an account with Google first, before even self-developed program can be installed on your own Android device
- Investigate the argument for both sides, Google saying it improves security of the platform, while developers say it restricts the freedom of everyone to run whatever program on one’s own phone?

Topic 3 - TPM Applications/Use

- Finde examples of common applications that require a TPM already today.
 - No Microsoft apps (e.g. Bitlocker)
- Examples: games anti-cheat
- Other high-value/expensive applications?
- What are their arguments that they absolutely need a TPM, which cannot be solved otherwise?

Topic 4 - Mobile Security

- In mobile/tablet devices there is no TPM
 - adding a dedicated chip is waaaay too costly
- How is a “secure” state enforced and assessed?
- How do apps with high-security needs, e.g. banking apps, ensure that the device is reasonable secure?
- What user interface problem exists in a mobile device with a limited screen UI?
 - Is it possible to solve it?

Topic 5 - Governmental Cloud

- Should our/your government use cloud services?
 - What is the situation in your country?
- How is responsible handling of citizen's data ensured?
 - “We signed a contract with them, so everything should be ok?”
- Have there been notable security incidents/scandals, where private data of citizens have been leaked?

Topic 6 - ...

- Your Topic?

- Individual work
 - Shared topic with someone – you can research/work together, but for the report delimit your topic and write your own text – no overlap!
- Do not use LLMs/ChatGPT/etc. to simply summarize material and copy&paste paragraphs together
- Document your findings in a report
 - ~4 pages
 - no specific style required (but use normal font size, margins, etc.)
- Upload as PDF (filename: Familyname-Title.pdf) on ECampus
- ...until **Sat Oct 25th 23:59**

Thank you! Questions?

Feedback welcome
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